

Predicting Transfer Market Value in Football using Football Manager 2024 Data

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Dedication

This dissertation is dedicated to my parents, Okunkpolor Charles and Okunkpolor Anthonia. Without your unwavering support, none of this would have been possible.

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Abstract

This study develops a transparent, data-driven framework for estimating the market value of professional football players by leveraging the structured player-attribute database of *Football Manager 2024* (FM24) supplemented with contractual, reputational and UEFA's league-strength information. Existing valuation approaches rely heavily on subjective judgement, crowd-sourced estimates, or limited performance metrics, resulting in inconsistent and often opaque valuation practices. To address this gap, a comprehensive dataset of over 13,000 European players was cleaned, transformed and enriched through domain-specific feature engineering encompassing ability, potential, age profiles, reputation, wage structure, contract terms, and a bespoke league-strength indicator derived from UEFA coefficients.

Three ensemble machine-learning models Random Forest, Gradient Boosting Regressor, and XGBoost were trained on the log-transformed FM24 midpoint valuations and evaluated using MAE, RMSE, R^2 and MAPE. All models demonstrated strong predictive performance (R^2 = 0.85–0.87; RMSE < £6M), with XGBoost achieving the strongest performance, reaching R^2 = 0.868 with an RMSE of £5.49M, outperforming both Random Forest and Gradient Boosting Regressor. SHAP analyses showed consistent feature hierarchies across models, with current ability, reputation, age-ability interactions, and potential ability emerging as the strongest valuation drivers. To enhance practical realism, predicted values were then adjusted to reflect known market patterns, including age-related depreciation and position-specific differences in demand, ensuring closer alignment with real transfer-market behaviour.

Comparison against Transfermarkt valuations showed expected differences, which reflect the subjective and context-dependent nature of public player valuations rather than model inaccuracy. Overall, the study demonstrates that ensemble learning combined with structured football-economics features can produce accurate, explainable and market-aligned player valuations, offering meaningful applications for recruitment, financial planning and decision support within modern football analytics.

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Abbreviations

1. **FM24** - *Football Manager 2024*
2. **GBR** - *Gradient Boosting Regressor*
3. **RF** - *Random Forest*
4. **XGB** - *XGBoost*
5. **MAE** - *Mean Absolute Error*
6. M**APE** - *Mean Absolute Percentage Error*
7. **RMSE** - *Root Mean Squared Error*
8. **R^2** - Coefficient of Determination
9. **UEFA** - *Union of European Football Associations*
10. **FIFA** - *Fédération Internationale de Football Association*
11. **M** - Million

1.0 Introduction

In recent years, football has evolved from a sport into a billion-dollar industry with billions of pounds exchanged every year in player transfers. The European football transfer market has become a key area of economic analysis, with transfer fees reflecting not only performances in game but also, brand value, reputation and commercial potential (Müller et al., 2017). However, despite the growth of data analytics in football, player valuations remain largely subjective, often driven by perception, reputation, and crowd judgement (Herm et al., 2014). This lack of transparency, objectivity, and a centrally recognised database of football data poses significant challenges for clubs, investors, and analysts seeking to understand or predict player market values.

Transfer fees are typically reported by platforms such as Transfermarkt (https://www.transfermarkt.co.uk/) which rely on crowd-sourced expertise rather than empirical modelling (Müller at al., 2017). To move towards more data-driven insights, researchers and clubs increasingly turn to machine learning methods to quantify what drives a player's worth. Models incorporating age, performance, contract duration and potential have demonstrated predictive utility, though their accuracy and interpretability vary across career stages, positions and modelling approaches (Khalife et al., 2025).

The primary research problem addressed in this study is a lack of a transparent, data-driven model capable of estimating player value using accessible, standardized data. Earlier valuation models often combined club-level financial information, subjective judgments, and a limited set of performance indicators (Dobson & Gerrard, 1999; Frick, 2007; Herm et al., 2014). These approaches often fail to capture the multitude of factors that can determine a transfer value, like injury history, release clauses present in contracts, the reputation of the league, these factors also influence a player's worth beyond raw performance. Existing research offers only partial solutions. Studies applying econometric and machine-learning approaches to predict footballers' market values often rely mainly on performance statistics, overlooking contractual, reputational, and contextual factors (McHale & Holmes, 2023). However, many existing approaches still lack domain-specific feature engineering grounded in football economics.

This study builds a machine-learning pipeline using Football Manager 2024 data to estimate player market values and compares these estimates to Transfermarkt valuations, incorporating additional contextual factors such as reputation, contract factors and league strength. The project also seeks to compare the performance of different ensemble models (Random Forest, Gradient Boosting, XGBoost) and identify which features most influence value predictions.

The hypothesis underpinning this research is that a model trained on FM24 attributes enhanced by contract factors, league strength and reputation features can produce accurate and explainable estimates of player market value, rivalling or improving upon crowd-based estimates. The study is motivated not only by academic interest but also by the practical implications for clubs and analysts in player recruitment, risk management, and investment valuation. To understand how this study fits into existing research, it is important to review how previous work has approached valuing football players. Academic research on football player valuation has evolved from simple economic models to complex, data-driven frameworks. Early studies such as Dobson and Gerrard (1999) applied hedonic regression models, where observable player attributes like age, position, and goals were treated as linear predictors of transfer fees. While foundational, these models assume additive relationships and struggled to represent nonlinear dynamics like the decline of value after a player's peak age or the impact of market hype.

The emergence of machine learning has enabled more flexible and accurate approaches. Müller et al., (2017) were able to demonstrate that random forest models can outperform crowd-based estimates in predicting transfer values for Bundesliga players, highlighting the importance of non-linear relationships and interaction effects. However, their model was trained on a single league and lacked interpretability.

Al-Asadi and Tasdemir (2022) extended this idea using *FIFA 20* game data comprising over 17,000 players. They compared linear regression, decision trees, and random forests, concluding that tree-based ensembles captured hidden patterns such as how age and potential interact more effectively than linear models. Yet, their dataset ignored real-world contractual and reputational variables, limiting external validity.

Behravan and Razavi (2020) introduced a hybrid PSO-tuned Support Vector Regression (SVR) model, first clustering players by position and then predicting value within each group. This two-stage approach improved accuracy to 74 %, underscoring the benefit of position-specific modelling. Still, the model did not account for financial constraints or release clauses, which can drastically alter transfer fees for example, Erling Haaland's transfer to Manchester City in 2022 was triggered via a €60m release clause despite a market valuation reported by *Transfermarkt* of over €150m at the time (Transfermarkt, 2025).

Emerging data sources such as Football Manager (Sports Interactive, 2024) offer structured, scout-informed player metrics including Current and Potential Ability that mirror real-world evaluations and can enhance modelling efforts. These datasets provide consistent, standardised representations of ability, potential, and reputation, addressing several limitations found in studies relying solely on match statistics or crowd-sourced information such as Transfermarkt.

2.0 Methods

2.1 Data Source and Description

The dataset was extracted from Football Manager 2024 (FM24) (Sports Interactive, 2024), a commercially licensed game available for purchase at https://www.sports-interactive.com/games/football-manager-2024. The game's database contains details of approximately 15,800 professional male players across more than 50 leagues. Player values in the FM database are typically expressed as intervals (e.g., *£50M–£70M*); therefore, the midpoint (*fm_value_mid*) was adopted as a consistent single-value estimate for model training and evaluation. The dataset also included players whose transfer values were marked as 'not for sale', for whom FM24 does not provide a transfer-value range, resulting in missing *fm_value_mid* values. These players were excluded from training and evaluation and were instead used only for final predictions once the model was fully trained. UEFA league coefficients (UEFA, 2025) were sourced directly from the official UEFA website and manually assigned to each European league in the dataset.

2.2 Data Preparation and Cleaning

All data preparation was performed in Python 3.12 using *pandas* and *NumPy*. To ensure comparability, only players registered in European leagues and aged 18 or older were retained, resulting in a cleaned dataset of about 13,400 players.

Custom preprocessing functions were developed to convert FM's text-based attributes into numeric form. Monetary strings such as "£50M", "£10K", and "£120k p/w," were parsed with parse_money, converting shorthand suffixes to full numeric values. Transfer-value ranges (e.g., "£50M–£65M") were processed by parse_transfer_value, which extracted minimum, maximum, and midpoint values; the midpoint (fm_value_mid) served as the dependent variable. Wages were standardised to weekly GBP values (parse_wage), while player height, weight, and contract dates were normalised using conversion helpers (parse_height_cm, parse_weight_kg, parse_date).

Contract duration was calculated from the signing date to the expiry date, and remaining contract length was measured relative to the FM24 snapshot date (1 July 2023). Additional variables were derived to capture performance and financial structure, including Potential Growth = Potential Ability – Current Ability**,** normalised ability and reputation scores, and injury-proneness indices. Release-clause fields were parsed numerically, with binary flags indicating whether clauses existed. To represent league competitiveness, a composite league strength indicator was constructed by combining official UEFA coefficients with within-dataset scaling factors.

Finally, player position data often listing multiple roles (e.g., "D/WB (R)") were simplified by retaining the first position and grouping players into four categories: Goalkeeper (GK)**,** Defender (DEF)**,** Midfielder (MID)**,** and Forward (FWD)**.** These categories were one-hot encoded (e.g., pos_GK, pos_DEF) to enable model compatibility.

2.3 Feature Engineering

Feature engineering was conducted to extract measurable indicators of player ability, contractual status, financial conditions, and contextual reputation that influence market valuation.

Derived features were organised into four main categories:

- Performance and Potential: normalised ability ratings (*CA_norm*, *PA_norm*), development measures (*Potential Growth*, *Potential Gap*), and injury-adjusted ability scores captured each player's technical quality and progression capacity.
- Contract and Financial Attributes: contract length, expiry, and wage information were transformed into metrics such as *Contract Ratio*, *Contract Decay*, and *Wage-to-League Ratio* to reflect negotiation leverage and economic context.
- Reputation and Competitiveness: Current Reputation (CR), World Reputation (WR) and variables like *Reputation Gap* (*CR – WR*) and a composite *League Strength* index combining UEFA coefficients with league-level scaling modelled both international reputation and league quality.
- Demographic and Interaction Factors: nonlinear and interaction effects were introduced through *Age²*, *Age × CA*, *Elite Flag*, and *Age-Over-Peak* variables to capture career-stage dynamics.

All continuous variables were scaled between 0 and 1, and outliers were retained to preserve genuine market phenomena.

Table 1 and Table 2 below give a breakdown of the features involved in the study

Attribute (FM24)	FM24 Minimum Value	FM24 Maximum Value
Current Ability	1	200
Potential Ability	1	200
Current Reputation	1	10,000
World Reputation	1	10,000
Injury Proneness	1	20

Table 1: Official Football Manager 2024 Attribute Scales for Key Variables (Note: These ranges reflect the internal attribute limits set within the Football Manager 2024 database and were not modified by the researcher.)

Feature Name	**Description**	**Purpose / Rationale**
Performance and Potential		
CA_norm	Normalised Current Ability (0–1)	Represents player's current skill level.
PA_norm	Normalised Potential Ability (0–1)	Shows player's development ceiling.
Potential Growth	Potential Ability – Current Ability	Measures development headroom.
Potential Gap	PA_norm – CA_norm	Captures relative growth capacity.
Potential Ratio	CA_norm / PA_norm	Indicates proximity to full potential.

CA_adj	Injury-adjusted ability score	Penalises players with higher injury proneness.
Injury Penalty	Weighted score combining injury risk and ability loss	Models depreciation from injury susceptibility.
Contract and Financial Attributes		
Contract Years Total	Duration of contract (signing to expiry)	Measures overall contract length.
Contract Years Remaining	Time left before contract expiry (as of July 2023)	Reflects transfer leverage and stability.
Contract Ratio	Remaining / total years	Indicates proportion of contract fulfilled.
Contract Decay	Exponential decay of remaining time	Models reduced bargaining power near expiry.
Contract Remaining²	Squared term for remaining years	Captures nonlinear decline in leverage.
Wage_log	Log-transformed weekly wage	Reduces skewness in financial distributions.
Wage-to-League Ratio	Wage divided by league competitiveness	Compares salary against market context.
Release Clauses	Monetary values for all clause types	Represents contractual exit conditions.
Has Release Clause Flags	Binary indicator for release clauses	Captures presence or absence of release terms.
Reputation and Competitive Context		
CR_norm	Normalised Current Reputation (0–1)	Indicates domestic reputation of Player.
WR_norm	Normalised World Reputation (0–1)	Reflects international player reputation.
Reputation Gap	WR_norm – CR_norm	Shows difference between global and domestic recognition.
League Strength	Combined UEFA coefficient × value scaling	Measures overall competition quality per league.
League Strength Scaled	Normalised version of league strength (0–1)	Ensures comparability across leagues.
Experience Strength	League strength × contract years	Approximates player exposure to competition.
Club Reputation Proxy	League strength × current reputation	Represents the prestige context of playing in a strong league
Consistency Proxy	Mean of CR_norm and WR_norm	Represents reliability and recognition stability.

Nationality-League Interaction	League strength × nationality factor	Captures regional familiarity or adaptation effects.
Demographic and Age Dynamics		
Age	Age of the player	Represents actual age of the player
Age_sq	Age squared	Models non-linear ageing effects.
Age_CA	Age × Current Ability	Represents synergy between experience and ability.
Age_over	Years above performance peak (27 years)	Measures post-peak decline.
Age_under	Years below peak	Represents growth potential pre-peak.
Age_over_sq	Squared term of years above peak	Models' acceleration of decline.
Elite Flag	1 if CA_norm ≥ 0.93 and CR_norm ≥ 0.93	Identifies top-tier elite players.
Age_over_elite	Age_over × Elite Flag	Captures slower decline among elite players.
Age_CA_ratio	Age / (CA_norm × 100)	Indicates age relative to ability level.
Elite Age Balance	Inverse weighting for elite age effects	Adjusts performance retention in older elites.
Target and Core Variables		
fm_value_mid	Midpoint of FM24 transfer-value range	Dependent variable for model training.
Wage_clean	Parsed weekly wage (numeric)	Standardised wage input for analysis.
is_not_for_sale	1 if player marked "Not for Sale"	Represents market unavailability status.
Position Groups (pos_GK, pos_DEF, pos_MID, pos_FWD)	One-hot encoded position categories	Allows model differentiation by playing role.

Table 2: Feature Dictionary: Definitions and Rationale for All Engineered Variables

2.4 Model Development

Three ensemble tree-based regression models were implemented using *scikit-learn* and *XGBoost* to predict player market value:

a. **Random Forest Regressor** (baseline)

The Random Forest model (Breiman, 2001) provided a robust baseline capable of handling multicollinearity and complex, non-linear feature interactions. It aggregates multiple decorrelated decision trees via bootstrap aggregation (*bagging*), reducing variance and overfitting risk.

b. **Gradient Boosting Regressor (GBR)**

Gradient Boosting (Friedman, 2001) was implemented to sequentially minimise residual errors from weak learners, thereby improving predictive precision relative to bagging methods.

c. **XGBoost Regressor**

The XGBoost (Chen & Guestrin, 2016) algorithm (short for *Extreme Gradient Boosting*) was adopted for its regularisation, parallelisation, and computational efficiency.

All models used the same pre-processed Feature Dictionary (Table 2) and the target variable fm_value_mid, log-transformed using *log1p* to reduce heteroskedasticity (Wooldridge,2016). Predicted values were subsequently converted back from the logarithmic scale to their original monetary units (£) for interpretation. An 80:20 train-validation split was used for model evaluation. Ensemble approaches were selected over deep-learning methods because the dataset is moderate-sized and primarily structured, favouring interpretable, tree-based models.

Performance differences across models reflected the trade-off between bias reduction (boosting) and variance control (bagging).

2.5 Post-Prediction Realism Adjustments

To ensure that the model's outputs aligned with realistic football-market dynamics, two post-processing adjustments were applied to the predicted transfer values. These corrections aimed to incorporate known behavioural patterns that are difficult for purely statistical models to capture.

1. **Age-based depreciation and youth premium:**
   Player value was adjusted for career stage using age-sensitive coefficients. A 3.5% annual depreciation was applied for each year beyond age 27 to reflect declining market appeal after peak performance (Lehmann & Schulze, 2008), while a 1.5% youth bonus per year below age 21 accounted for developmental potential and resale value.
2. **Positional multipliers:**
   Empirically derived coefficients Goalkeepers (0.55), Defenders (0.75), Midfielders (1.00), and Forwards (1.00) were applied to the adjusted predictions to reflect position-specific market demand. This ensured, for example, that goalkeeper valuations remained within realistic bounds while forwards were appropriately valued higher.

These post-prediction calibrations improved the interpretability and face validity of the model outputs, bringing predicted transfer values closer to observed market behaviour.

2.6 Model Evaluation

Model performance was evaluated on the validation set using MAE, RMSE, R^2 and MAPE to assess both absolute and relative prediction accuracy. All metrics were calculated on unlogged predictions to preserve interpretability in monetary terms. MAE was used as the primary metric, with RMSE and R^2 providing complementary indicators of error magnitude and explanatory power.

2.7 Explainability and Feature Interpretation

To enhance transparency and interpretability, model outputs were analysed using SHAP (SHapley Additive exPlanations), a game-theoretic approach that attributes each feature's contribution to individual predictions (Lundberg & Lee, 2017; Molnar, 2022).

Two levels of interpretation were performed:

- **Global explainability** identified the most influential predictors across the entire dataset. Summary plots showed that *Current Ability (CA_norm)*, *Potential Ability (PA_norm)*, *Wage_log*, *League Strength*, and Age_*Contract_Ratio* exerted the largest positive influence on predicted market value.

- **Local explainability**: Force plots for individual cases such as Thibaut Courtois decomposed each predicted value into additive feature effects relative to the model's baseline expectation, illustrating how specific attributes shaped valuation outcomes.

This interpretability analysis verified that the ensemble models captured meaningful economic and performance relationships rather than over-fitting artefacts, thereby improving confidence in the model's internal logic and alignment with real-world football-market behaviour.

3.0 Results

3.1 Overview

This chapter presents the results of three ensemble regression models Random Forest (RF), Gradient Boosting Regressor (GBR) and XGBoost (XGB) trained to predict footballers' market values using the *Football Manager 2024* dataset.

Performance is assessed using standard regression metrics (MAE, RMSE, R^2 and MAPE). Additional subsections describe the distribution of predictions, feature-importance patterns, SHAP-based explainability, and realism adjustments that contextualise the estimated transfer values.

3.2 Model Performance

Table 3 summarises the validation performance of all three models. All models achieved strong performance, with $R^2 \geq 0.83$ and RMSE below £6 million, indicating that they captured most of the variance in player valuations and delivered high predictive accuracy. Among them, XGBoost delivered the lowest error values and achieved a marginally higher R^2 than the other two models.

Model	MAE (£)	RMSE (£)	R^2	MAPE (%)
Random Forest	1,305,557	5,885,936	0.848	75.26%
Gradient Boosting	1,233,176	5,661,471	0.86	71.28%
XGBoost	1,218,616	5,494,139	0.868	71.50%

Table 3. Model Performance Comparison on Validation Set

3.3 Release Clauses and Model Estimates

To demonstrate that the model recognises contractual clauses that affect transfer values. The predicted transfer values for the players were compared with a sample of ten players with release clauses.

Name	Club	Release Clause (£M)	Random Forest Predicted Market Value	Xgboost Predicted Market Value	Gradient Boosting Regressor Predicted Market Value
David Alaba	R. Madrid	£729M	£43.0M	£37.8M	£48.4M
Thibaut Courtois	R. Madrid	£600M	£62.1M	£66.2M	£65.1M
Robert Lewandowski	Barcelona	£429M	£28.5M	£36.5M	£18.6M

Marc-André ter Stegen	Barcelona	£429M	£49.6M	£50.4M	£45.4M
Antoine Griezmann	A. Madrid	£429M	£33.6M	£42.4M	£41.3M
İlkay Gündoğan	Barcelona	£343M	£20.5M	£32.4M	£19.7M
William Carvalho	Real Hispalis	£103M	£14.0M	£12.3M	£14.9M
Gerard Moreno	Villarreal	£86M	£26.9M	£25.9M	£30.4M
Iago Aspas	Vigo	£43M	£4.9M	£5.2M	£6.3M
Marcos Acuña	Sevilla	£35M	£8.5M	£9.1M	£10.3M

Table 4. Comparison of Predicted Market Values and Release Clauses for Selected Players

3.4 Top-11 Predicted Values after Realism Adjustment

Following realism corrections (age decay 3.5 % per year after 27 and positional multipliers: GK-0.55, DEF-0.75, MID/FWD-1.00), the predicted transfer values aligned with realistic transfer dynamics.

Table 5 presents the top 11 highest-valued players according to the models, alongside their corresponding Transfermarkt valuations (Transfermarkt, 2025). The values gotten from Transfermarkt are from 2023 except Mohamed Salah which was in late 2022.

In Table 5, a positive value in the Average Model Prediction - Transfermarkt (£M) column indicates that the models predicted a higher market value than Transfermarkt, whereas a negative value indicates that the models predicted a lower value.

Player	**Club**	**Random Forest Prediction (£M)**	**XGBoost Prediction (£M)**	**Gradient Boosting Prediction (£M)**	**Closest Transfermarkt Value (£M)**	**Average Model Prediction minus Transfermarkt (£M)**
Antoine Griezmann	Atlético Madrid	33.6	42.4	41.3	22	17.1
Casemiro	Manchester United	37	41.7	58	35	10.57
David Alaba	Real Madrid	43	37.8	48.4	44	-0.93
Erling Haaland	Manchester City	185.7	192	214	158	39.23
Kevin De Bruyne	Manchester City	83.2	109.6	65.8	61	25.2

Marc-André ter Stegen	Barcelona	49.6	50.4	45.4	35	13.47
Mohamed Salah	Liverpool	119.6	125	111.2	71	47.6
Robert Lewandowski	Barcelona	28.5	36.5	18.6	35	-7.13
Thibaut Courtois	Real Madrid	62.1	66.2	65.1	40	24.47
Virgil van Dijk	Liverpool	89.8	93.2	71.4	40	44.8
Gerard Moreno	Villarreal	26.9	25.9	30.4	27	0.73

Table 5: Comparison of Top 10 Predicted Market Values Across Models and Transfermarkt Values (note: transfermarkt figures have been converted from euros to pounds)

3.5 Feature Importance

Global feature-importance plots (Figures 1–3) show that CA_norm**,** CR_norm**,** and PA_norm are the most influential predictors across all ensemble models. Age-related variables such as age_CA_ratio**,** age_under**,** and age_over_sq also appear consistently important. Contextual and economic features including Wage_log**,** contract_decay**,** club_reputation_proxy**,** and uefa_points exhibit moderate contributions. The overall ranking of features is broadly similar across the Random Forest, XGBoost, and Gradient Boosting models. These findings are consistent with prior research (Müller et al., 2017; Al-Asadi & Tasdemir, 2022).

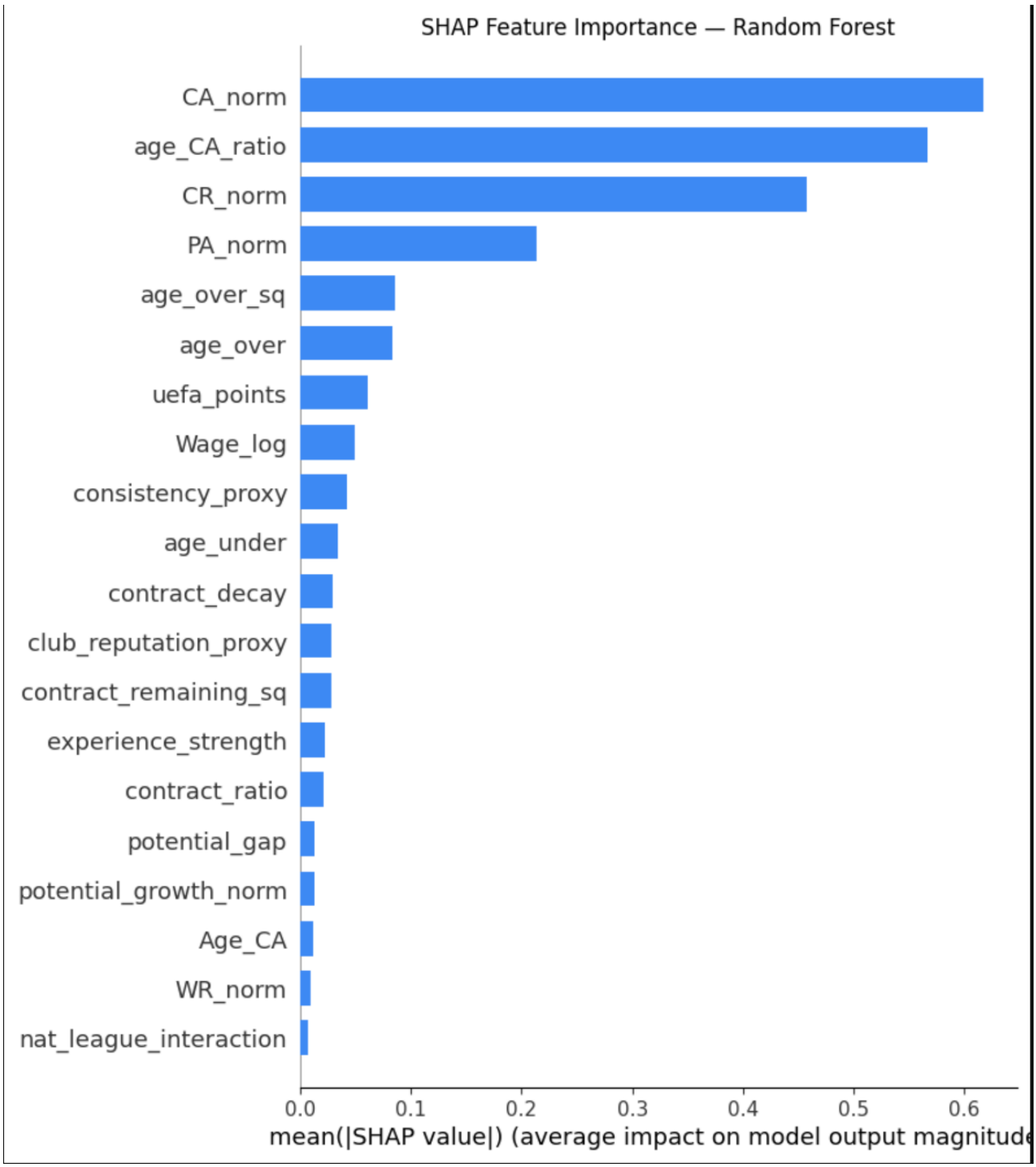


Figure 1: Global feature importance plot for the Random Forest model

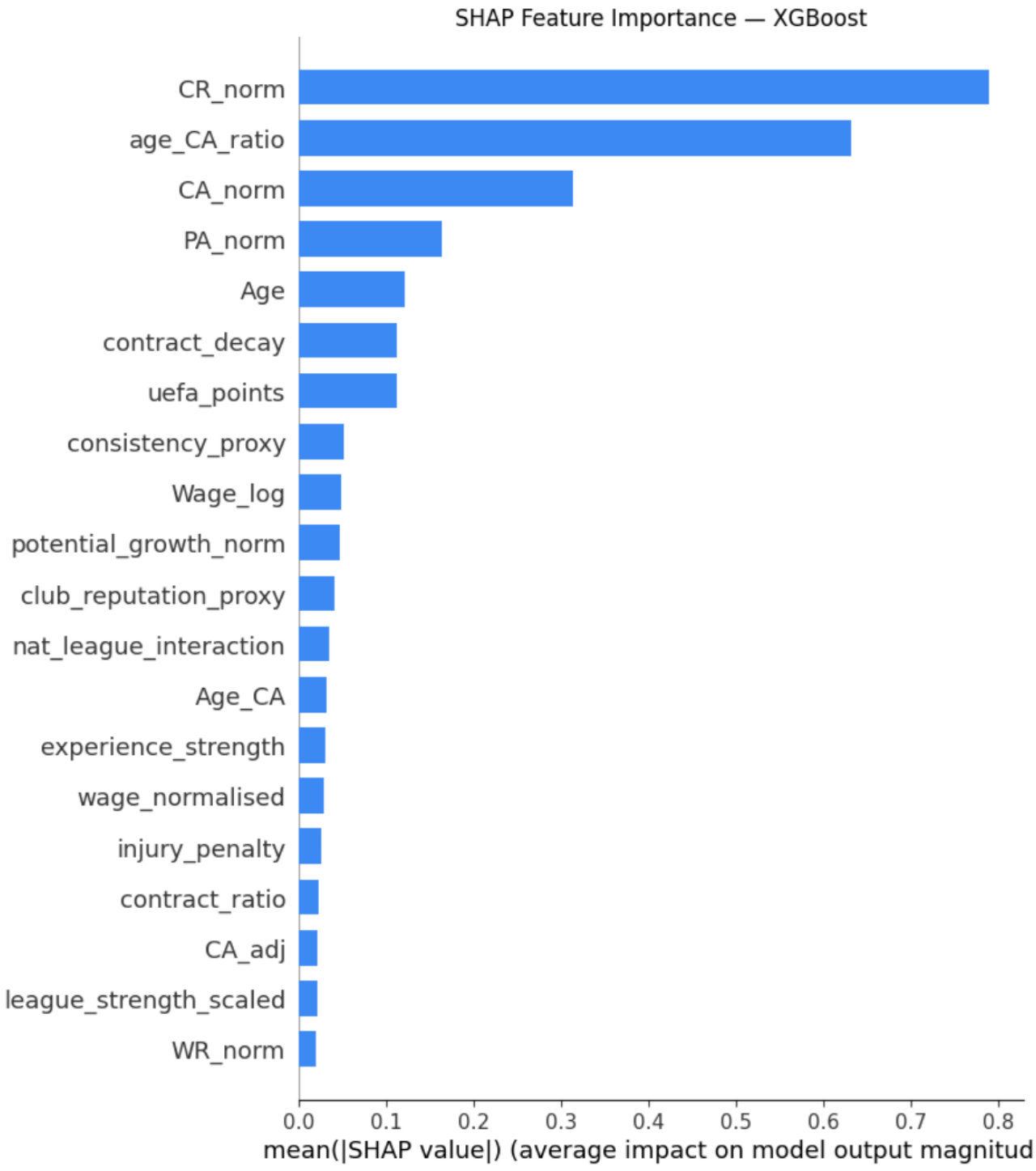


Figure 2: Global feature importance plot for the Xgboost model

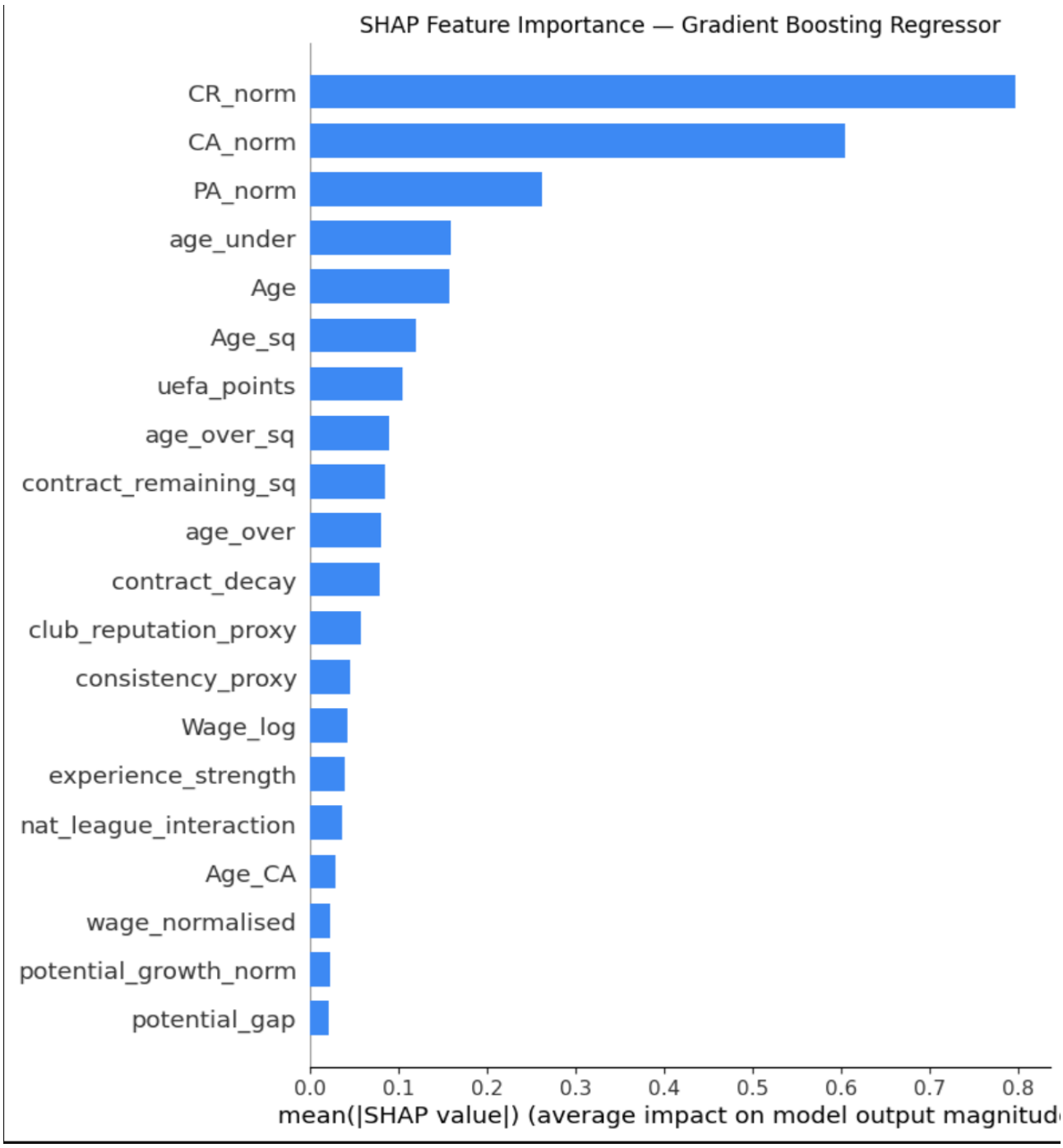


Figure 3: Global feature importance plot for the gradient boosting regressor model

3.6 Local Explainability

SHAP analyses were used to generate player-level feature contribution tables for Thibaut Courtois and Erling Haaland across all three ensemble models.

For Courtois, the top features included CA_norm, CR_norm, PA_norm, and age_CA_ratio, while variables such as Wage_log, contract_decay, and club_reputation_proxy also appeared among the main contributors.

For Haaland, all models identified CR_norm, CA_norm, and PA_norm as primary features, alongside age_CA_ratio, uefa_points, and contract_decay.

Feature rankings varied slightly between the Random Forest, XGBoost, and Gradient Boosting models but remained broadly consistent in highlighting similar predictors.

Random Forest			Xgboost			Gradient Boosting Regressor		
Feature	**Value**	**SHAP_Impact**	**Feature**	**Value**	**SHAP_Impact**	**Feature**	**Value**	**SHAP_Impact**
CA_norm	0.905	2.05408	CR_norm	0.905	2.6024	CR_norm	0.905	2.938244
CR_norm	0.905	1.79151	age_CA_ratio	0.3425	1.1266	CA_norm	0.905	1.970173
age_CA_ratio	0.342541	1.10711	CA_norm	0.905	0.9685	PA_norm	0.905	0.783271
PA_norm	0.905	0.8616	PA_norm	0.905	0.7117	club_reputation_proxy	0.748419	0.325217
consistency_proxy	0.88	0.1928	club_reputation_proxy	0.7484	0.159	consistency_proxy	0.88	0.154036
Wage_log	12.42922	0.15553	Wage_log	12.429	0.1481	potential_growth_norm	0	0.1209
club_reputation_proxy	0.748419	0.08128	contract_decay	0.0499	0.1403	Wage_log	12.42922	0.115267
experience_strength	4.027929	0.02352	potential_growth_norm	0	0.1084	contract_remaining_sq	8.987684	0.105376
contract_decay	0.049889	0.02337	consistency_proxy	0.88	0.1072	potential_gap	0	0.103927
WR_norm	0.855	0.02193	Age_CA	28.055	0.092	contract_decay	0.049889	0.100845

Table 6: SHAP Feature Contributions for Thibaut Courtois: Random Forest, XGBoost, and Gradient Boosting

Random Forest			Xgboost			Gradient Boosting Regressor		
Feature	**Value**	**SHAP_Impact**	**Feature**	**Value**	**SHAP_Impact**	**Feature**	**Value**	**SHAP_Impact**
CA_norm	0.94	2.04633	CR_norm	0.94	2.576985	CR_norm	0.94	2.485183
CR_norm	0.94	1.72953	age_CA_ratio	0.234043	1.296663	CA_norm	0.94	1.752624
age_CA_ratio	0.234043	1.24025	CA_norm	0.94	0.971122	PA_norm	0.975	0.921243
PA_norm	0.975	0.87625	PA_norm	0.975	0.69183	club_reputation_proxy	0.94	0.320251
consistency_proxy	0.92	0.1768	uefa_points	96.227	0.235532	uefa_points	96.227	0.195771
Wage_log	12.765691	0.16046	club_reputation_proxy	0.94	0.157355	Age	22	0.137755
club_reputation_proxy	0.94	0.11312	contract_decay	0.018366	0.156036	contract_decay	0.018366	0.129559
uefa_points	96.227	0.06723	Wage_log	12.765691	0.152982	contract_remaining_sq	15.978105	0.125884
age_over_sq	0	0.05216	consistency_proxy	0.92	0.08255	age_under	5	0.117922
age_over	0	0.04757	WR_norm	0.9	0.07829	Age_sq	484	0.115924

Table 7: SHAP Feature Contributions for Erling Haaland : Random Forest, XGBoost, and Gradient Boosting

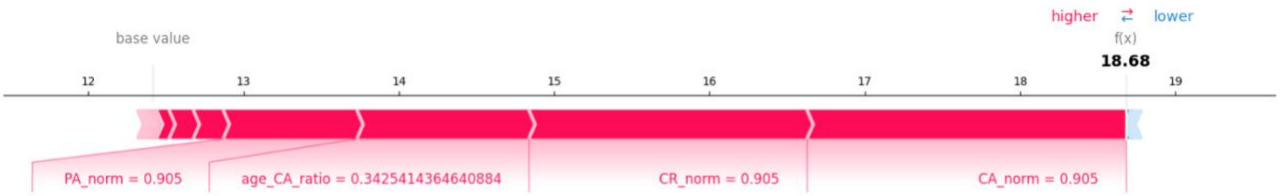


Figure 4: Random Forest SHAP force plot for Thibaut Courtois showing feature contributions to predicted value.

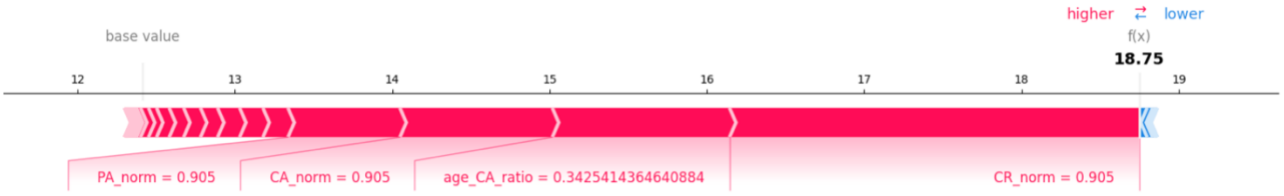


Figure 5: Xgboost SHAP force plot for Thibaut Courtois showing feature contributions to predicted value.

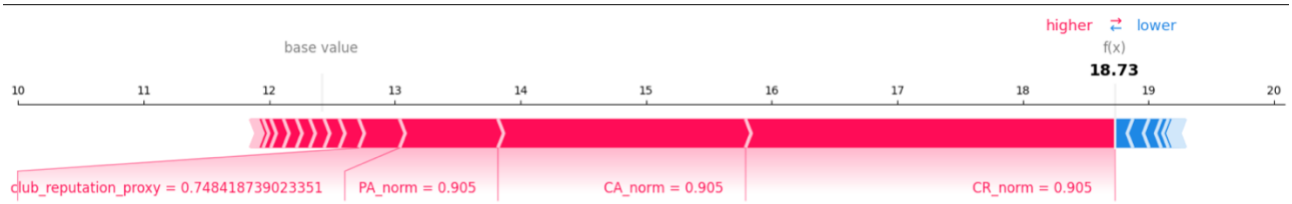


Figure 6: Gradient Boosting Regressor SHAP force plot for Thibaut Courtois showing feature contributions to predicted value.

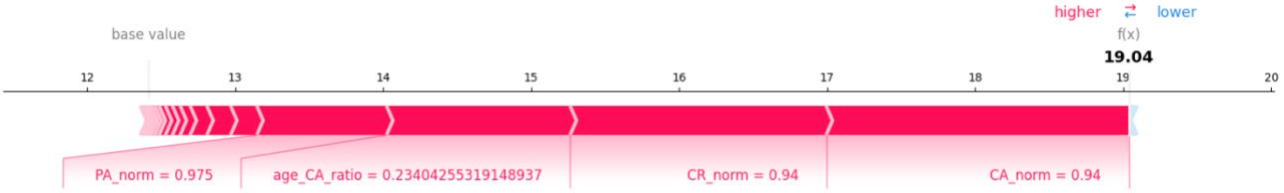


Figure 7: Random Forest SHAP force plot for Erling Haaland showing feature contributions to predicted value.

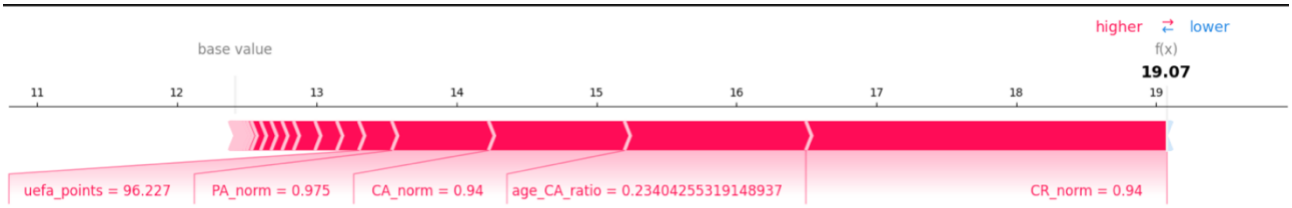


Figure 8: Xgboost SHAP force plot for Erling Haaland showing feature contributions to predicted value.

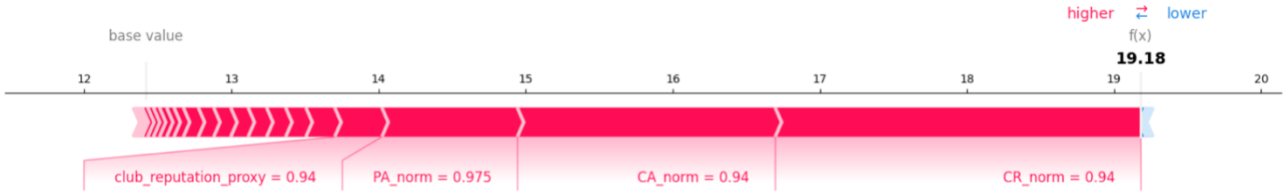


Figure 9: Gradient Boosting Regressor SHAP force plot for Erling Haaland showing feature contributions to predicted value.

4.0 Discussion

This chapter interprets the results of the ensemble models used to predict professional footballers' market values from Football Manager 2024 (FM24). Three tree-based algorithms Random Forest, Gradient Boosting Regressor (GBR) and XGBoost were compared in predictive performance, interpretability, and alignment with established valuation research.

4.1 Comparative Model Performance

As shown in Table 3**,** all three ensemble methods achieved strong predictive accuracy capturing the complex, non-linear structure of football-player valuation. The Random Forest baseline achieved an RMSE of approximately £5.89 million and $R^2 = 0.848$, while both boosting algorithms outperformed it: GBR (RMSE ≈ £5.66 million, $R^2 = 0.860$) and XGBoost (RMSE ≈ £5.49 million, $R^2 = 0.868$). This aligns with prior work on football-player valuation (Müller et al., 2017), which applies multilevel regression modelling to estimate market value and highlights the broader shift towards data-driven analytical approaches in this domain. XGBoost's performance advantage stems from its built-in regularisation and efficiency mechanisms, which help control model complexity and reduce over-fitting (Chen & Guestrin, 2016). This aligns with findings by Al-Asadi and Tasdemir (2022), who observed that advanced ensemble methods outperform simpler linear approaches in football-player valuation tasks.

4.2 Model Valuations and Comparison with *Transfermarkt* Valuations

The Top-11 ranking in Table 5 illustrates that the three ensemble models generated consistent and economically plausible estimates of elite player valuations. Across all models, Erling Haaland emerged as the most valuable player, with predicted values ranging from £185.7 million (Random Forest) to £214 million (Gradient Boosting), compared with a Transfermarkt valuation of roughly £158 million in June 2023. Similarly, Kevin De Bruyne's predicted value ranged from £65.8 million (GBR) to £109.6 million (XGBoost), compared with a Transfermarkt valuation of £61 million *(Transfermarkt*, 2025), while Mohamed Salah's estimates spanned £111–£125 million versus a listed Transfermarkt value of £71 million.

The comparison also revealed cases where model predictions diverged from *Transfermarkt's* published figures in both directions. For example, Antoine Griezmann's predicted values ranged from £33.6–£42.4 million versus £22 million on *Transfermarkt* (*Transfermarkt*, 2025), while Robert Lewandowski's predicted values (£18.6–£36.5 million) were close to but slightly below his £35 million listing on *Transfermarkt* (*Transfermarkt*, 2025). These outcomes suggest that the models strongly captured the influence of technical ability and reputation, while applying moderate discounts for ageing forwards.

In contrast, Thibaut Courtois and Virgil van Dijk were valued higher by the models (£62.1–£66.2 million) and (£71.4–£93.2 million), respectively compared with their £40 million *Transfermarkt* valuations (*Transfermarkt*, 2025). These cases indicate that the models weighted consistent performance, reputation and contract duration alongside age and position-adjusted factors, producing higher valuations for Courtois and van Dijk than the £40 million Transfermarkt valuation (Transfermarkt, 2025). The differences highlight how model outputs are grounded in structured player attributes and calibrated depreciation rules, whereas Transfermarkt valuations also incorporate wider market sentiment, positional norms and negotiation dynamics.

Transfermarkt, while widely regarded as a credible public reference, is not a definitive source of true market prices it reflects community-moderated estimates rather than verified transaction data (Herm et al. 2014). Numerous real-world transfers diverge substantially from their listed valuations, underscoring that price discovery in football is dynamic and often shaped by negotiation, timing, and club-specific financial pressures (Frick, 2007). Therefore, the observed differences between predicted and Transfermarkt values do not necessarily indicate model inaccuracy but rather expose the subjective and context-dependent nature of player valuation. By producing a coherent player hierarchy that aligns with general market logic yet acknowledges deviations from Transfermarkt's figures, the ensemble models demonstrate both predictive validity and practical interpretive value within the uncertain economics of football trading (Frick, 2007).

4.3 Release Clauses and Model Estimates

Table 4 compares each model's predicted market values with official release clauses for selected top-tier players. Across all cases, release clauses were vastly inflated relative to model predictions for instance, Courtois (£600 million clause) and Alaba (£729 million) were valued by the models between £40 million and £65 million. This confirms that clauses function primarily as contractual safeguards rather than genuine indicators of market value (Frick, 2007). They are designed to deter transfers and preserve club bargaining power rather than to mirror an achievable selling price.

XGBoost and Gradient Boosting tended to generate slightly higher valuations than Random Forest, consistent with studies showing that boosting-based approaches can offer improved predictive accuracy in football player valuation (Yigit et al., 2020). For older players such as Lewandowski and Gündoğan, predicted values were far below their contractual clauses, illustrating how static clause figures fail to adjust to on-field depreciation.

Overall, these results emphasise that release clauses are strategic and legally protective instruments, while the models estimate practical market value.

4.4 Global Feature Influence

The SHAP feature-importance plots for all three ensemble models (Figures 1–3) reveal a broadly consistent structure but with subtle variations in emphasis. In the Random Forest model (Figure 1), Current Ability (CA_norm) dominates, followed by age_CA_ratio and Current Reputation (CR_norm), indicating that a player's technical quality, efficiency relative to age, and individual reputation within the game are the strongest drivers of predicted value. The XGBoost model (Figure 2) instead places *CR_norm* first, ahead of *age_CA_ratio* and *CA_norm*, showing a slightly greater sensitivity to reputation effects for the Gradient Boosting Regressor (Figure 3), Current Reputation (CR_norm) emerges as the dominant feature, followed by Current Ability (CA_norm). Potential Ability (PA_norm) and age-related terms (such as age_under) also contribute meaningfully, showing emphasis on established reputation and proven technical quality while still rewarding youth and long-term potential.

Across models, *Wage_log* and contract-related variables such as *contract_decay* and *contract_ratio* maintain secondary influence, capturing bargaining power and time-to-expiry dynamics. Across all three models, Potential Ability (PA_norm) emerges as a consistently important predictor. However, growth-based potential metrics such as potential_gap and potential_growth_norm, as well as

experience_strength, contribute very little, indicating that the models focused more on a player's absolute potential than on their developmental trajectory or experience.

These SHAP rankings confirm established economic relationships in player valuation. Player ability and reputation remain the strongest determinants of value, consistent with Müller et al. (2017), and Behravan and Razavi (2020). The prominence of age_CA_ratio and PA_norm highlights the recognised market premium for younger players with greater potential (Müller et al., 2017; Peeters, 2018). Meanwhile, wage and contract-related variables reflect bargaining power and time-to-expiry dynamics, aligning with the contractual valuation frameworks of Frick (2007) and Behravan and Razavi (2020).

4.5 Individual SHAP Feature Analysis

The player-level SHAP analyses (Tables 6–7; Figures 4–9) illustrate how each model explained individual market valuations through combinations of ability, reputation, and contextual factors.
To illustrate how the models interpret value for different player profiles, two contrasting case studies were examined: Thibaut Courtois, representing an experienced goalkeeper in a position that is historically under-analysed in valuation research, and Erling Haaland, a globally recognised forward whose market value is widely understood and provides a clear benchmark for high-profile attackers. These examples allow us to assess how the models handle both a specialised positional role and an elite outfield player.

For Thibaut Courtois (Real Madrid)**,** ability-driven features dominated across all models. In the Random Forest model (Table 6; Figure 4), *CA_norm* (2.05), *CR_norm* (1.79) and *age_CA_ratio* (1.11) produced the highest SHAP impacts, indicating that Courtois's predicted value was shaped primarily by sustained performance and reputation rather than potential upside. The moderate contribution from *PA_norm* (0.86) reinforces this, reflecting Courtois's maturity and positional stability as an experienced goalkeeper. In the XGBoost model (Table 6; Figure 5), CR_norm (2.60) was the strongest contributor, followed by age_CA_ratio (1.13) and CA_norm (0.97). This indicates that Courtois's valuation was most strongly influenced by his high Current Reputation and elite ability, with the age-adjusted ability term further reflecting his sustained performance level despite advancing age.

The Gradient Boosting Regressor (Table 6; Figure 6) ranked *CR_norm* (2.94) and *CA_norm* (1.97) highest, with *PA_norm* (0.78) and *club_reputation_proxy* (0.33) further enhancing the estimate. Collectively, the models attributed Courtois's high valuation to elite performance consistency and the reputational boost of representing Real Madrid aligning with Frick (2007) who identified club prestige and sustained ability as major determinants of value for established players.

For Erling Haaland (Manchester City)**,** SHAP attributions were more strongly influenced by offensive potential and youth. In the Random Forest model (Table 7; Figure 7), *CA_norm* (2.05) and *CR_norm* (1.73) ranked highest, followed by *age_CA_ratio* (1.24) and *PA_norm* (0.88), suggesting a balance between proven ability and forward-looking development potential. The XGBoost model (Table 7; Figure 8) shifted emphasis slightly CR_norm (2.58) and age_CA_ratio (1.30) produced the strongest effects, with CA_norm (0.97) and PA_norm (0.69) also contributing. This pattern aligns with established evidence that younger players with higher potential and good reputation tend to receive higher market valuations, particularly in attacking roles (Herm et al., 2014; Behravan & Razavi, 2020).The Gradient Boosting Regressor (Table 7; Figure 9) displayed a similar pattern, with *CR_norm* (2.49), *CA_norm* (1.75), and *PA_norm* (0.92) jointly explaining most of the predicted valuation, while *club_reputation_proxy* (0.32) and *UEFA_points* (0.20) further elevated his predicted worth.

Together, these results demonstrate that both players' valuations emerged from coherent but role-dependent logic. Courtois's value was driven by reputation stability and technical consistency traits typically associated with elite goalkeepers while Haaland's valuation reflected youth, attacking productivity, and highly competitive reputation at a major club. The contrast highlights the adaptability of ensemble SHAP explanations across positional contexts, consistent with Müller et al. (2017), who observed that forwards command higher premiums due to potential and visibility, while goalkeepers are valued more for established performance. The emphasis on youth and potential for Haaland also aligns with Behravan and Razavi (2020), who found that players with higher potential and stronger reputational indicators tend to be systematically valued more highly.

Although FM24 provides structured data, it remains a simulated database rather than an official public source. Organisations such as FIFA, UEFA, and domestic leagues publish only aggregated transfer statistics rather than detailed datasets containing individual player contracts, wages, technical attributes, or verified market valuations (FIFA, 2024).

Second, Transfermarkt valuations were used as a comparative benchmark, but these figures are community-informed estimates rather than verified transaction prices. Real-world transfers often deviate from these published values due to negotiation dynamics, timing, club-specific financial pressures and strategic behaviour during transfer windows.

A key direction for future work is to replace community-estimated Transfermarkt valuations with verified transactional transfer fees sourced from audited club financial reports, FIFA's Transfer Matching System (TMS), or reputable media disclosures. Using real transfer amounts would allow the model to learn from actual market behaviour, capturing negotiation outcomes, timing effects, club financial constraints, and strategic pricing mechanisms that public estimates cannot fully represent. This would strengthen the model's validity and reduce reliance on subjective or crowd-sourced valuation data.

5.0 Conclusion

This study set out to predict professional footballers' market values using Football Manager 2024 data combined with contextual attributes and three ensemble learning models (Random Forest, Gradient Boosting Regressor, and XGBoost). The results showed that these models effectively captured the non-linear and multi-factor structure of player valuation, achieving strong predictive accuracy ($R^2 \approx 0.85–0.87$) and offering transparent explanations through SHAP.

Overall, the findings demonstrate that ensemble learning can replicate key economic principles of player pricing by integrating ability, potential, reputation, and contract characteristics. XGBoost delivered the strongest performance, consistent with prior work showing the advantages of boosting-based methods. The comparison with Transfermarkt values and release clauses further illustrated that public valuations include subjective and strategic components, whereas model predictions respond more directly to underlying quantifiable attributes.

Feature-level and player-specific SHAP analyses confirmed that the models adapted logically to positional roles and career stages, strengthening confidence in their interpretability and practical relevance.

In conclusion, this research provides a transparent, data-driven approach to football player valuation that aligns with established sports-economics theory.

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